

MagicCode Super-Resolution Software FAQ and Troubleshooting

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1. General

What does MagicCode Super-Resolution do?

It upscales images or GPU frames. v2.2.0 public modes are SPATIAL_SPEED_MODE, SPATIAL_BALANCED_MODE, TEMPORAL_SPEED_MODE, and TEMPORAL_BALANCED_MODE.

Does the software upload my images or videos?

No. Technical reporting does not upload image or video content. See the Privacy Policy and Data Reporting Notice (unchanged by this technical release).

What scale factors are supported?

Spatial: [1.0, 8.0]. Temporal: (1.0, 8.0]. Actual availability depends on backend and model. API support is not a claim that every scale is device-run.

2. Temporal FAQ

Is MC_Enable still available?

No. v2.2.0 removed MC_Enable and mc_enable.h. Use mc_nscaler_enable; temporal frames attach a temporal_frame_t on mc_nscaler_input_frame_t.frame.

If I omit masks, do I have a semantic transparency mask?

No. The library derives an approximate mask when inputs are missing. Explicit textures win per channel. FSR-family paths bind a zero transparency texture when omitted and do not generate semantic transparency from color.a.

Which backends and platforms support temporal?

Android Vulkan/OpenGL ES, Apple Metal, Windows Vulkan / Direct3D 11 / OpenGL 4.3. Linux Vulkan temporal builds from source (the shipped Linux archive lib/linux/libmagic_sr.a is CPU spatial only). CPU temporal is unavailable.

Is there a Linux build?

Yes. lib/linux/libmagic_sr.a provides the x86_64 CPU spatial path (AVX2/FMA). Rebuild with tools/build_linux_lib.sh and link -lm -lpthread. GPU temporal on Linux uses the Vulkan backend

built from source; tools/build_linux_vulkan_itest.sh is a headless reference integration. CPU temporal is unavailable.

Temporal Super-Resolution

Temporal super-resolution runs through the nscaler session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable. v2.2.0 replaced the v2.0.0 MC_Init / MC_Process / MC_Uninit entry and removed the v2.1.0 MC_Enable helper (mc_enable.h is deleted). A NULL *handle on mc_nscaler_enable creates a session with defaults (2x SPATIAL_SPEED) and processes the first frame.

Modes

Value	Enumerator	Role
0	SPATIAL_SPEED_MODE	Spatial, throughput first
1	SPATIAL_BALANCED_MODE	Spatial, quality/cost trade-off
2	TEMPORAL_SPEED_MODE	Temporal, throughput first
3	TEMPORAL_BALANCED_MODE	Temporal, canonical FSR-family path

- Temporal scaler_factor is (1, 8]; 1.0 itself is rejected. Spatial remains [1.0, 8.0].
- TEMPORAL_SPEED_MODE on Metal, Vulkan, and OpenGLS currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family temporal path (same family as Balanced, not a separate customer API).
- TEMPORAL_BALANCED_MODE uses the canonical FSR-family path on every GPU temporal backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLS do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run temporal modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- mc_nscaler_input_frame_t / mc_nscaler_output_frame_t: caller-owned color and output GPU resources.
- mc_nscaler_input_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).

- `gpu_context` at create (`mc_nscaler_control SET_PARAM`, or the first `mc_nscaler_enable`): Vulkan requires `physical_device + device`; D3D11 requires `device`; Metal device may be `NULL`; GL/GLES use the current context (library never makes current).
- Vulkan temporal requires `mc_nscaler_input_frame_t.command_buffer` to be a recording `VkCommandBuffer`. Metal `command_buffer` is optional. D3D11 / GL / GLES leave it `NULL`.
- Create-stage `depth_reversed` / `depth_infinite` / `hdr_color` on `ctrl_param_t` (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- `motion_vector_scale_x/y` is the only MV numeric conversion. (0,0) defaults to (`input_width`, `input_height`). Mixed zero is rejected.
- Depth is GPU device depth in `[0, 1]`. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both temporal modes enable depth+MV history rejection by default.
- Diagnostic only: `MAGICSR_TAAU_AUTO_MASK=0` and `MAGICSR_TAAU_HIST_REJECT=0` can turn those defaults off. They are not a customer configuration path.

3. Initialization problems

- Model not found: `SetModelPath/SetModelDir`, packaging, permissions.
- Width/height out of range: 64..4032.
- Scale out of range: spatial `[1,8]`, temporal `(1,8]`.
- Backend unavailable: CPU temporal, or GPU temporal not in this build.
- Vulkan/D3D11 GPU create failed: missing `gpu_context.device` (and Vulkan `physical_device`).

4. Processing problems

- `MC_ERROR_TEMPORAL_NULL_FRAME`: `mc_nscaler_input_frame_t.frame` is `NULL`.
- Missing color/output/depth/motion handles.
- Vulkan `command_buffer` `NULL` or layout `UNDEFINED`.
- Ghosting: `reset_history` on hard cuts; check MV `current→previous` and +Y down jitter.
- Wrong output size: `QUERY_STATUS` and confirm `scaler_factor`.

5. Unity / Unreal

- `DllNotFoundException`: place the native plugin under `Assets/Plugins/Android/arm64-v8a/`.

- UE missing library: lib/android and lib/ios libmagic_sr.a (the v2.1.0 enable archive no longer exists).
- Plugin Enable cannot feed temporal extras; use Native session API for temporal.

6. Diagnostic checklist

- mc_nscaler_version should report v2.2.0.
- Record backend, input type, alg_mode, scale, width, height, model path.
- Record API return values and output_status_params_t.error_code.
- For temporal, confirm temporal_status_valid after a successful process.